

TABLE 2
RISK FACTORS IN THE TURBINE CONTROLLER DESIGN AND THEIR MITIGATION

Identified Risk	Type	Severity	Occurance	Detection	RPN	Mitigation Measure	New RPN
Dead battery, CPU will not work	Tech	7	6	10	420	Provide alternate power source from wind turbine	70
No battery connected, Ineffective CPU and free running of wind turbine	Tech	9	3	10	270	Provide alternate power source and connect to dummy when battery is not detected	30
Dummy load disconnected will cause free running of turbine	Tech	8	3	6	144	In absence of dummy load, activate break circuit	54
Short circuit in battery	Tech	9	3	5	135	Isolate battery and divert power to dummy load	75
Short circuit in dummy load	Tech	8	3	5	120	Isolate dummy load and activate break circuit	75
Strong wind resulting in high voltage	Tech	6	6	5	180	When turbine speed crosses a set threshold, activate brake circuit	45
Short circuit in both battery and dummy load	Tech	9	1	5	45	No action required	45
Power controller relay failure	Tech	9	3	9	243	By default connect to dummy load. Test proper operation at regular interval	48
Loose connection in power cables will overheat and burn terminals	Tech	6	6	6	216	Isolate terminals from main controller and put a overheating sensor	36
CPU failure	Tech	9	3	10	270	Provide a hooter with delay. Reset hooter timer by CPU	81
ADC failure	Tech	3	2	7	42	No action required	42
Wi-Fi connection failure	Tech	4	5	8	160	Provide check in server to raise notification in the event of long reporting delay	80
Scarcity of NodeMCU module	Sup	8	3	3	72	No action required	72
Scarcity of ADC multiplexer	Sup	8	2	3	48	No action required	48
Scarcity of relay	Sup	8	1	3	24	No action required	24
Price escalation of components	Eco	3	7	3	63	No action required	63
Competition from other manufacturers	Eco	3	5	3	45	No action required	45
Low demand of small wind turbines	Eco	3	4	3	36	No action required	36
Low demand of smart controller	Eco	3	5	3	45	No action required	45

case of any high current flow and prevent overheating and fire risks in house. In product design also, we often provide a ‘fuse.’ These are intentional weak points in the design that break down during any unplanned situation and prevent further damage to the system.

One must remember that fuse is a disruptive solution. Too much use of fuses is not good. As far as possible, one should use other innovative approaches to reduce damage and use a fuse only as the last resort.

Alarms and warnings are notifications for human intervention during emergency conditions. Such devices improve detection of faults. When the product is not able to

cope with certain unnatural conditions, it can give warning to the user to take some action. One must remember the importance of the signifier that we discussed in our first article.

Alarms and warnings must give clear information on what the user should do to deal with the emergency. Control panel or the interface needs to be suitably designed to eliminate any ambiguity between the warning and the desired action.

Example

Let us examine our charge controller for various types of risks. The problems can happen due to missing connections with the external parts.

We can also have situations of some module failing. As our production volume is expected to be small, we also have a risk of some of the modules being phased out by the respective manufacturer as well as price escalation.

On the economic front, there can be risk of competition. There is also the risk of low market demand, which may adversely affect the business viability.

In addition, we may get risk from the environmental conditions. Gusty winds may cause the wind turbine to generate higher power and voltage. If these are not controlled, it may damage the controller as well as the battery and alternator.